Muddaka Prasanth

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Summary

Software Developer with hands-on experience in MERN stack, backend engineering, and AI-driven applications. Built projects involving Neural Networks, KNN classification, LLM integration, RAG pipelines, and AI Agents using Python, NumPy, React.js, Node.js, and Express.js, with strong foundations in Data Structures and Algorithms.

Experience

Automation & QA Intern — StarWorks Technologies **Sep 2025**
Automation Tester (Selenium, Java, JIRA) *Remote Vizag,AP*

- Developed an **AI-powered Agent** to assist users and automate web navigation and interactions within the application.
- Designed agent workflows to handle user queries and trigger automated actions, improving usability and reducing manual effort.
- Contributed to automation testing using **Selenium WebDriver (Java)** for regression and functional testing.
- Performed UI and functional testing, identifying defects and improving application reliability.
- Tracked and managed issues using **JIRA** with detailed bug reports and resolution tracking.

Projects

Handwritten Digit Recognition System | *Python, NumPy, Matplotlib, Neural Networks, Flask* **Live** | **GitHub** **April 2026**

- Built **Neural Network from scratch** using NumPy with forward and backward propagation, and gradient descent.
- Trained and evaluated the model on MNIST dataset with preprocessing, normalization, and accuracy analysis.
- Developed a **Canvas UI** for capturing 28×28 handwritten input and real-time predictions.
- Deployed model inference APIs using cloud platforms (**AWS/Heroku**) for scalable and reliable predictions.

AI Agent Platform | *TypeScript, Node.js, Express.js, Langbase, Gemini API* **Live** | **GitHub** **March 2026**

- Built and deployed serverless **AI Agents** using **LLMs** with memory and context handling via Langbase.
- Implemented **RAG (Retrieval-Augmented Generation)** using **Cohere embeddings** for semantic search and vector-based retrieval.
- Integrated **Gemini API** and developed backend APIs with Node.js and Express.js for real-time processing.
- Designed multi-step agent workflows with **Prompts** and context optimization for accurate responses.

Crypto Wallet Dashboard | *React.js, Web3.js, Blockchain APIs* **Live** | **GitHub** **Jun 2025**

- Built a React.js dashboard to display crypto wallet activity using live blockchain API integration as part of Google WOWs Hackathon.
- Fetched and rendered real-time wallet balances and transaction history with sub-2s latency using **Web3.js** on Ethereum testnets.
- Enhanced UI responsiveness and usability with component-based design and optimized state management.

Breast Cancer Classification System | *Python, NumPy, Matplotlib, KNN, Machine Learning* **GitHub** **2025**

- Implemented the **K-Nearest Neighbors (KNN)** algorithm completely from scratch using only **NumPy**, without relying on Scikit-learn ML implementations.
- Trained and evaluated the model on the **Breast Cancer Wisconsin dataset** containing 500+ tumor samples for binary classification of malignant and benign cases.
- Built custom logic for **Euclidean distance calculation**, majority voting, normalization, train-test splitting, and accuracy evaluation from the ground up.
- Achieved over **95% classification accuracy** through feature preprocessing and optimized K-value experimentation.
- Visualized feature distributions, decision behavior, and classification outputs using **Matplotlib** to analyze model performance and prediction trends.

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, SQL
Frontend: React.js, HTML, CSS, Bootstrap, Responsive Design, Canvas API
Backend: Node.js, Express.js, Flask, REST APIs
AI / ML: Neural Networks, LLMs, AI Agents, RAG, Prompt Engineering, Embeddings
Databases: MongoDB, MySQL, PostgreSQL, Supabase
Cloud & Deployment: Heroku, Vercel, Serverless Architecture
Tools: Git, GitHub, Postman, VS Code, IntelliJ, UNIX CLI
Cloud & DevOps: Azure (Basics), Docker, Kubernetes, CI/CD Concepts(Jenkins) Basics
DSA: Trees, Graphs, Recursion, Dynamic Programming (300+ LeetCode, 70+ CodeStudio)

Open Source & Research Contributions

Zooniverse Contributor | *Citizen Science, Research Data Classification* **2026 – Present**

- Completed **500+ scientific classifications** across Zooniverse citizen-science projects involving image analysis and research dataset annotation.
- Contributed to projects such as **Beluga Bits** and **Chimp&See**, assisting researchers in wildlife identification and behavioral data collection.
- Collaborated with global volunteer-driven scientific initiatives focused on large-scale research data processing and classification workflows.
- Supported real-world research efforts by analyzing visual datasets and improving annotation quality for environmental and biological studies.

Hackathons & Achievements

Google WOW Hackathon – Top 30 Finalist | *Blockchain, Web3, React.js* **2025**

- Secured a position among the **Top 30 teams** in the Google WOW Hackathon by developing a blockchain-powered crypto wallet dashboard.
- Built a responsive **React.js** application integrating **Web3.js** and blockchain APIs for live wallet balance and transaction visualization.
- Implemented real-time blockchain data fetching with optimized state management, achieving **sub-2s response latency**.
- Collaborated in a competitive hackathon environment to design and deploy a scalable prototype within limited development timelines.

Green Guardian – Weed Detection System | *Python, Machine Learning, Computer Vision, Flask* Live **2026**

- Developed an AI-powered **weed detection system** for smart agriculture applications during a college hackathon competition.
- Secured **3rd Prize** by building a machine learning model capable of identifying weeds from agricultural field images for precision farming support.
- Processed and analyzed **1000+ agricultural image samples** using image preprocessing and classification workflows to improve detection reliability.
- Designed and deployed an interactive dashboard for real-time prediction visualization and agricultural monitoring workflows.
- Collaborated in a fast-paced team environment to design, train, test, and present a working prototype within strict hackathon timelines.

Education

Bachelor of Technology in Information Technology **2023 – 2027**
Pragati Engineering College *Surampalem, Andhra Pradesh — CGPA: 7.65*

Intermediate (MPC) **2019 – 2021**
Sri Chaitanya Junior College *Kakinada, Andhra Pradesh — Score: 823*